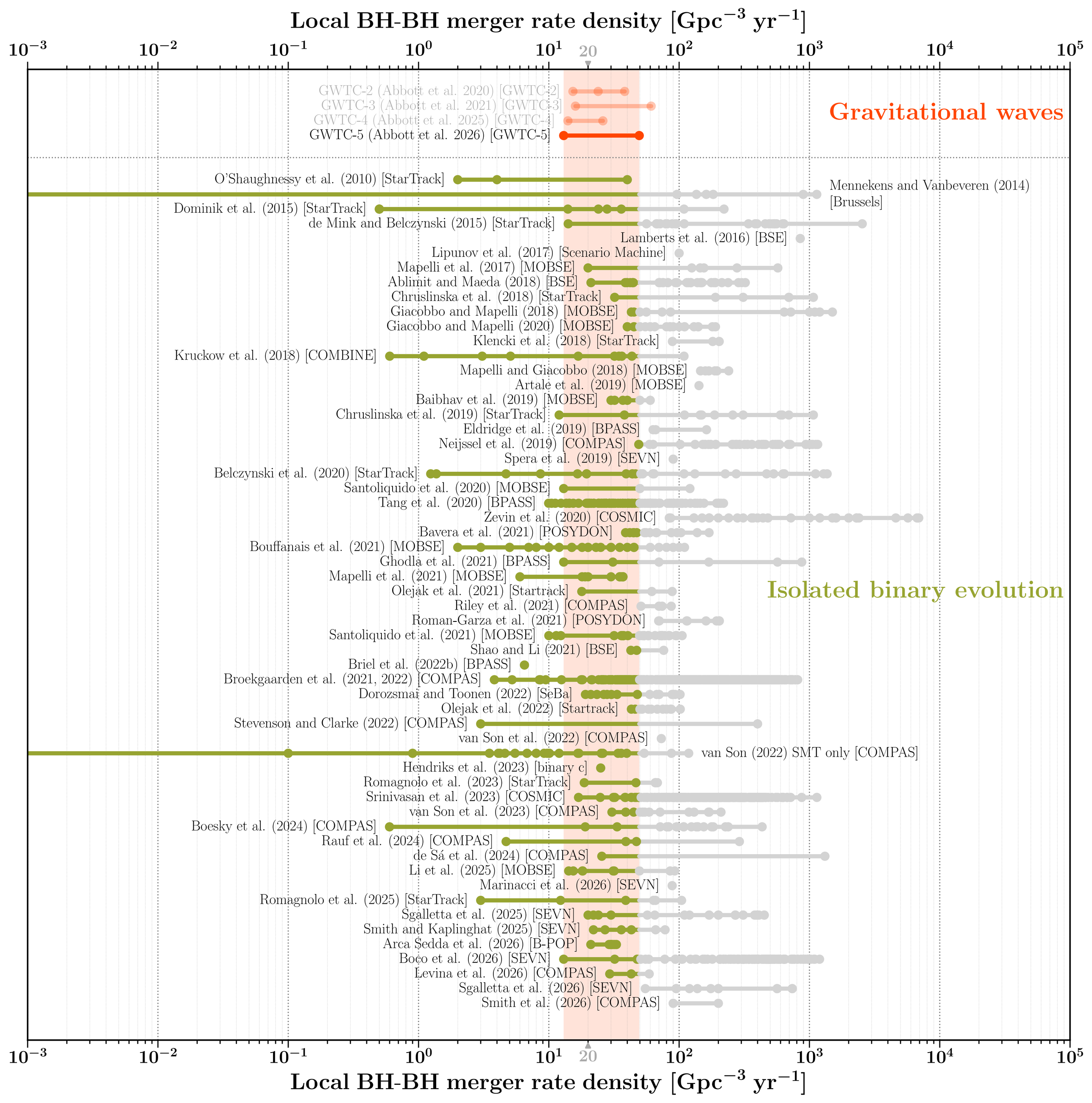

Local BH-BH merger rate density [Gpc^{-3} yr^{-1}]
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
20
10^{2}
10^{3}
10^{4}
10^{5}
Gravitational waves
GWTC-2 (Abbott et al. 2020) [GWTC-2]
GWTC-3 (Abbott et al. 2021) [GWTC-3]
GWTC-4 (Abbott et al. 2025) [GWTC-4]
GWTC-5 (Abbott et al. 2026) [GWTC-5]
Isolated binary evolution
O'Shaughnessy et al. (2010) [StarTrack]
Mennekens and Vanbeveren (2014) [Brussels]
Dominik et al. (2015) [StarTrack]
de Mink and Belczynski (2015) [StarTrack]
Lamberts et al. (2016) [BSE]
Lipunov et al. (2017) [Scenario Machine]
Mapelli et al. (2017) [MOBSE]
Ablimit and Maeda (2018) [BSE]
Chruslinska et al. (2018) [StarTrack]
Giacobbo and Mapelli (2018) [MOBSE]
Giacobbo and Mapelli (2020) [MOBSE]
Klencki et al. (2018) [StarTrack]
Kruckow et al. (2018) [COMBINE]
Mapelli and Giacobbo (2018) [MOBSE]
Artale et al. (2019) [MOBSE]
Baibhav et al. (2019) [MOBSE]
Chruslinska et al. (2019) [StarTrack]
Eldridge et al. (2019) [BPASS]
Neijssel et al. (2019) [COMPAS]
Spera et al. (2019) [SEVN]
Belczynski et al. (2020) [StarTrack]
Santoliquido et al. (2020) [MOBSE]
Tang et al. (2020) [BPASS]
Zevin et al. (2020) [COSMIC]
Bavera et al. (2021) [POSYDON]
Bouffanais et al. (2021) [MOBSE]
Ghodla et al. (2021) [BPASS]
Mapelli et al. (2021) [MOBSE]
Olejak et al. (2021) [Startrack]
Riley et al. (2021) [COMPAS]
Roman-Garza et al. (2021) [POSYDON]
Santoliquido et al. (2021) [MOBSE]
Shao and Li (2021) [BSE]
Briel et al. (2022b) [BPASS]
Broekgaarden et al. (2021, 2022) [COMPAS]
Dorozsmai and Toonen (2022) [SeBa]
Olejak et al. (2022) [Startrack]
Stevenson and Clarke (2022) [COMPAS]
van Son et al. (2022) [COMPAS]
van Son (2022) SMT only [COMPAS]
Hendriks et al. (2023) [binary c]
Romagnolo et al. (2023) [StarTrack]
Srinivasan et al. (2023) [COSMIC]
van Son et al. (2023) [COMPAS]
Boesky et al. (2024) [COMPAS]
Rauf et al. (2024) [COMPAS]
de Sá et al. (2024) [COMPAS]
Li et al. (2025) [MOBSE]
Marinacci et al. (2026) [SEVN]
Romagnolo et al. (2025) [StarTrack]
Sgalletta et al. (2025) [SEVN]
Smith and Kaplinghat (2025) [SEVN]
Arca Sedda et al. (2026) [B-POP]
Boco et al. (2026) [SEVN]
Levina et al. (2026) [COMPAS]
Sgalletta et al. (2026) [SEVN]
Smith et al. (2026) [COMPAS]
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
20
10^{2}
10^{3}
10^{4}
10^{5}
Local BH-BH merger rate density [Gpc^{-3} yr^{-1}]